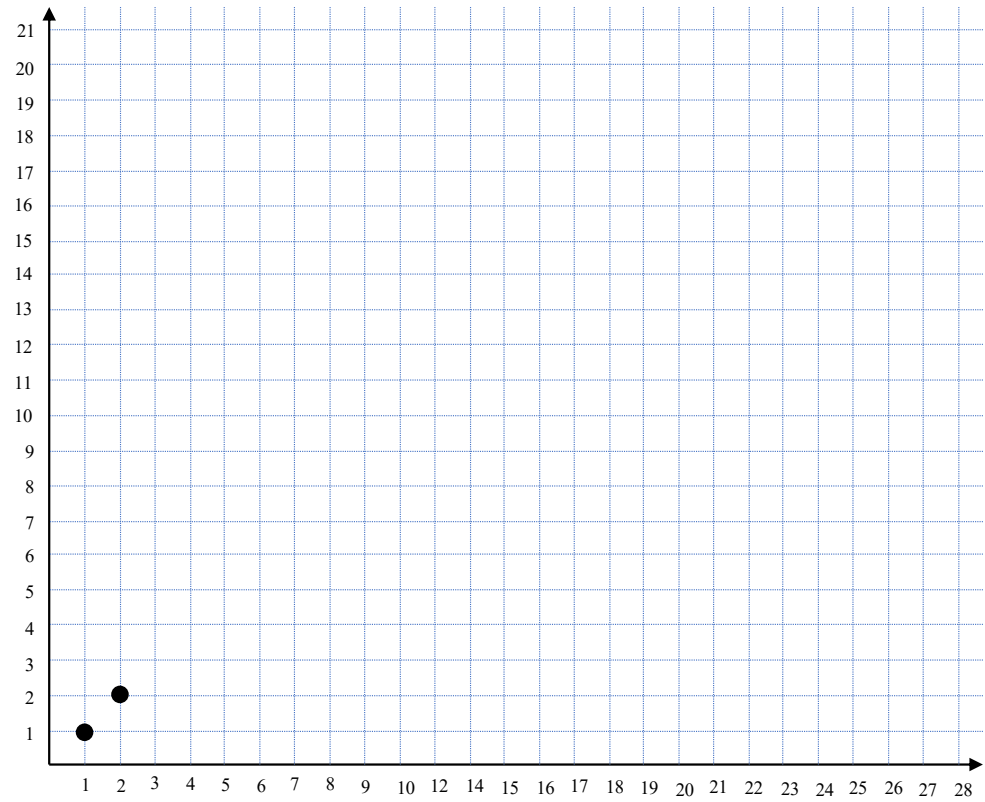


Linear Regression

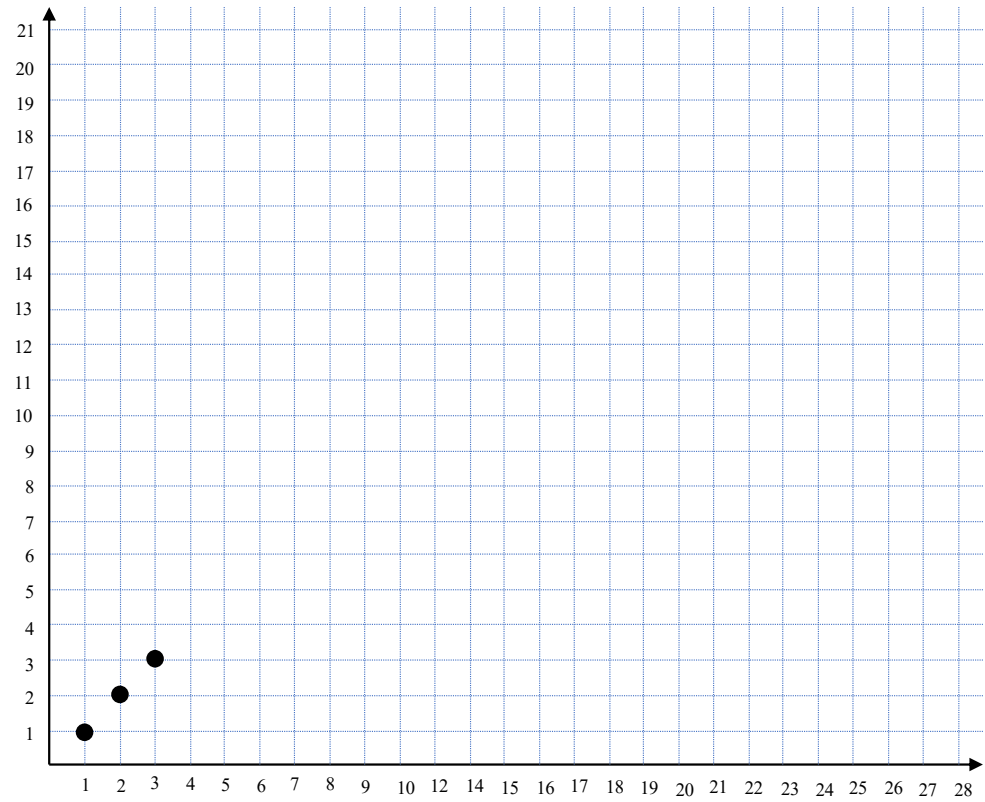
$y = f(x)$



x

x	$y = f(x) = x^1$
1	1
2	2

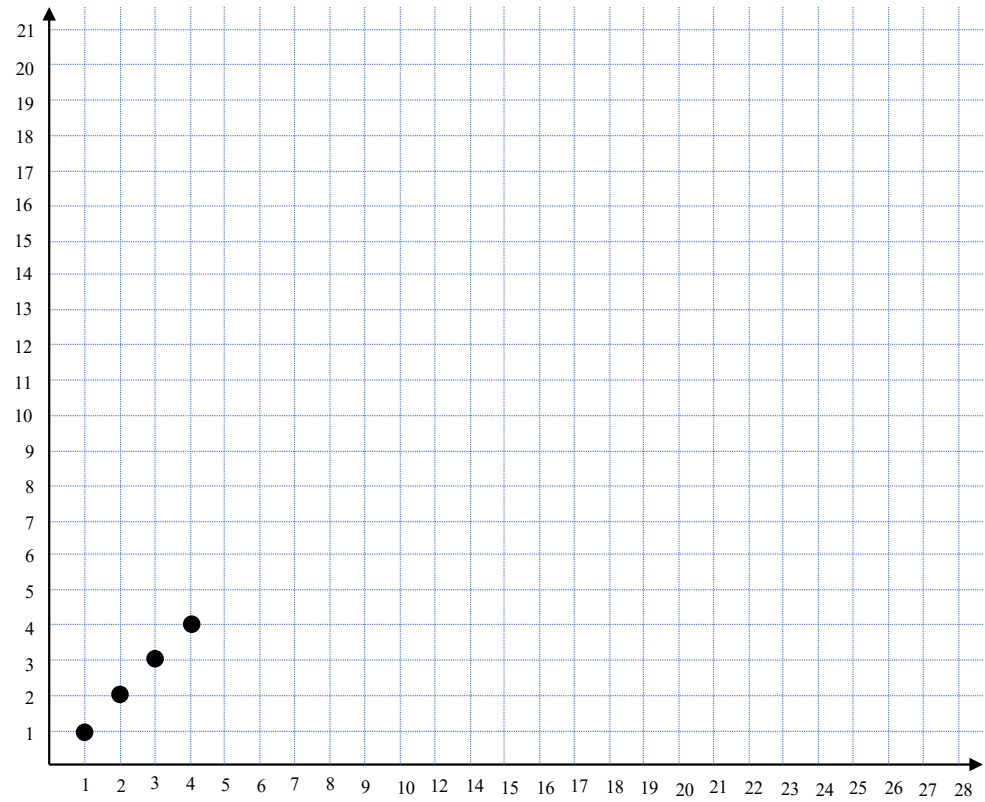
$y = f(x)$



x

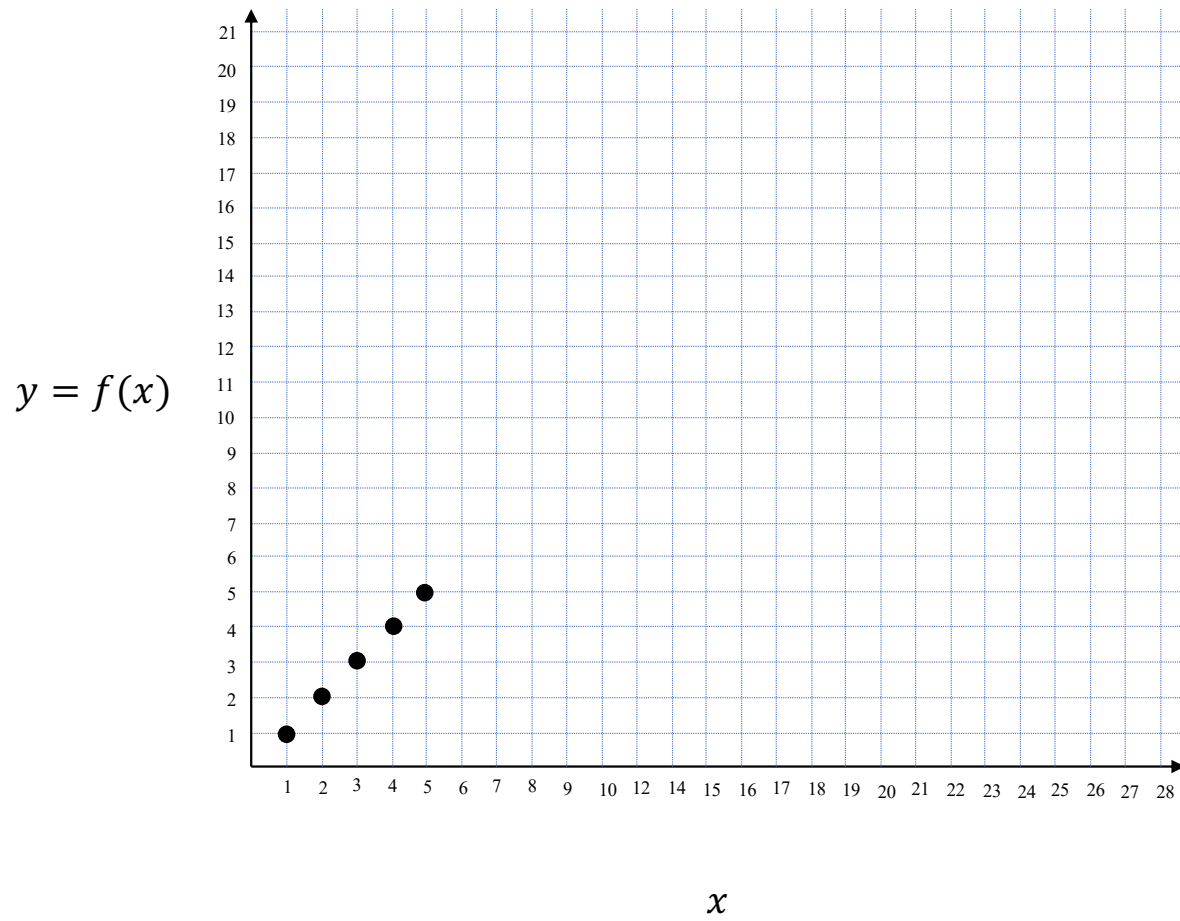
x	$y = f(x) = x^1$
1	1
2	2
3	3

$y = f(x)$

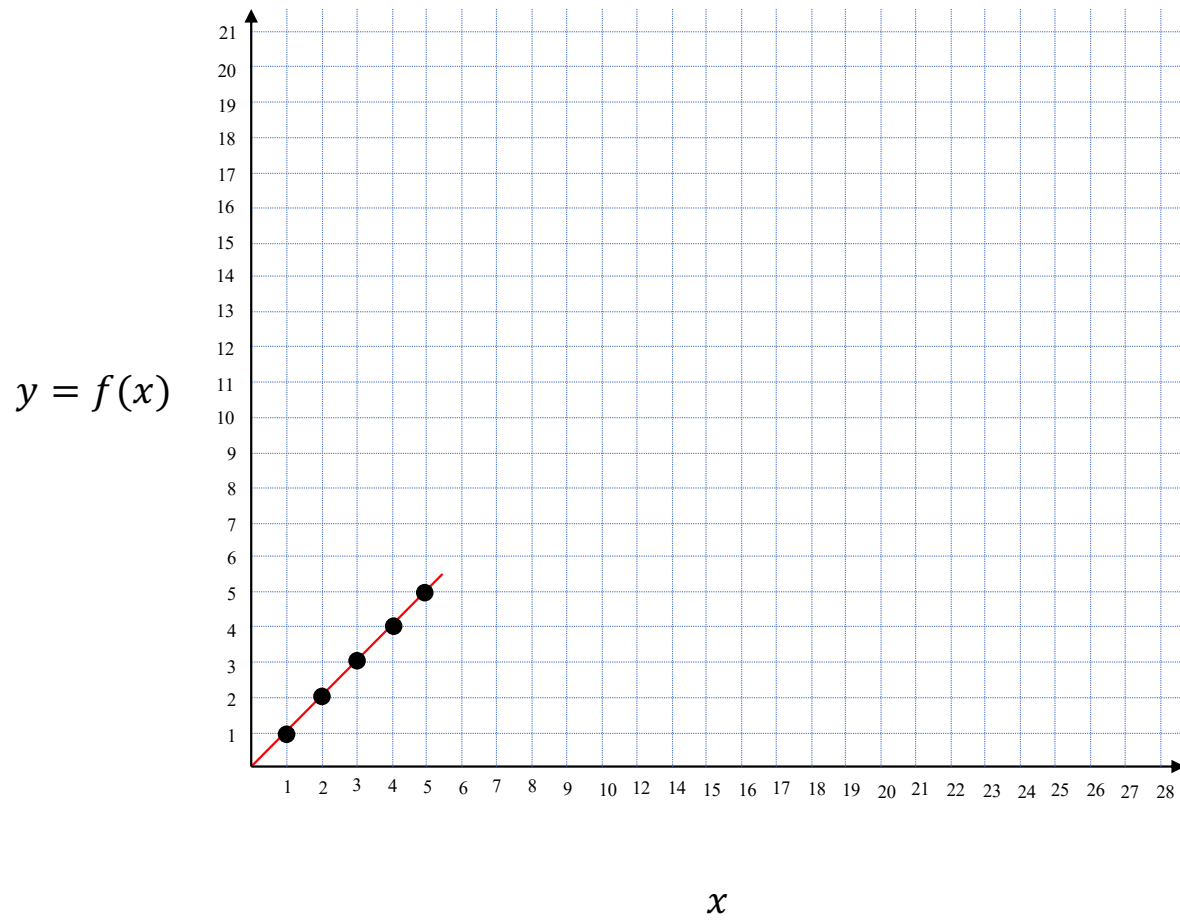


x

x	$y = f(x) = x^1$
1	1
2	2
3	3
4	4

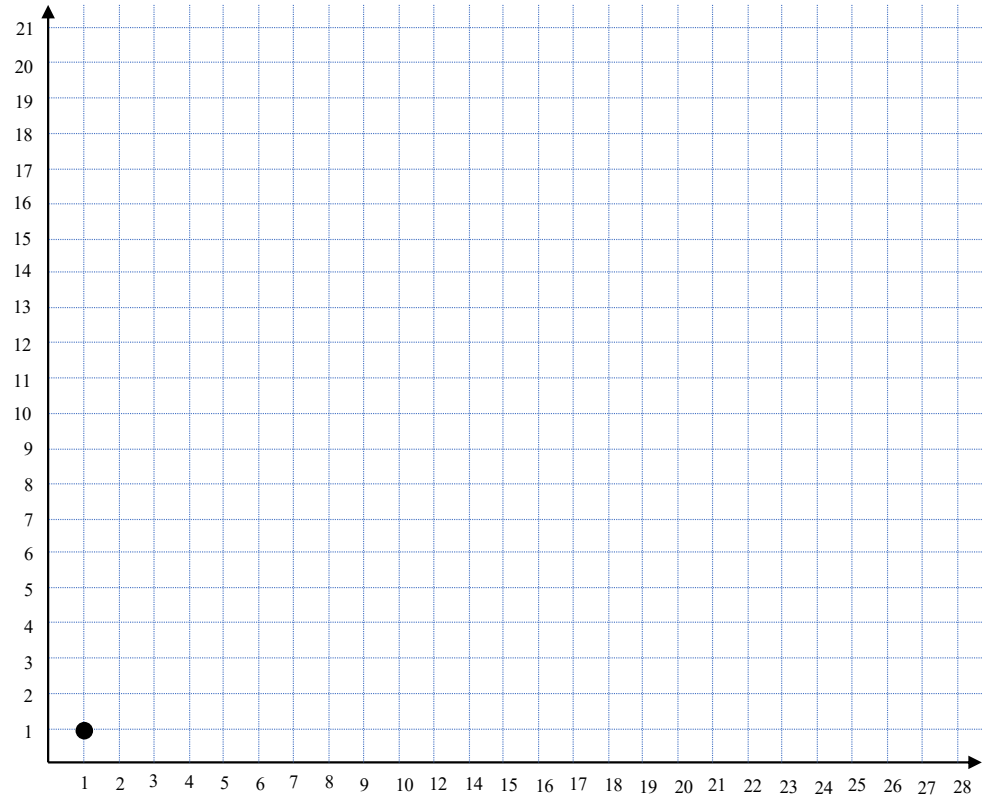


x	$y = f(x) = x^1$
1	1
2	2
3	3
4	4
5	5



x	$y = f(x) = x^1$
1	1
2	2
3	3
4	4
5	5

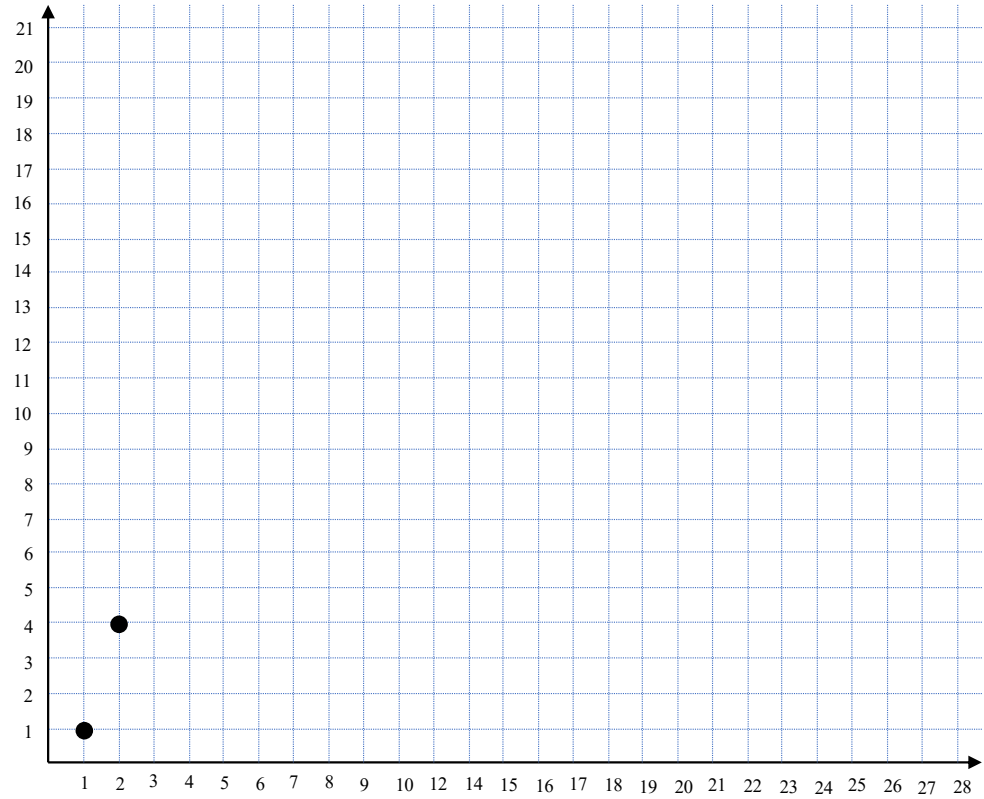
$y = f(x)$



x

x	$y = f(x) = x^2$
1	$1^2 = 1$

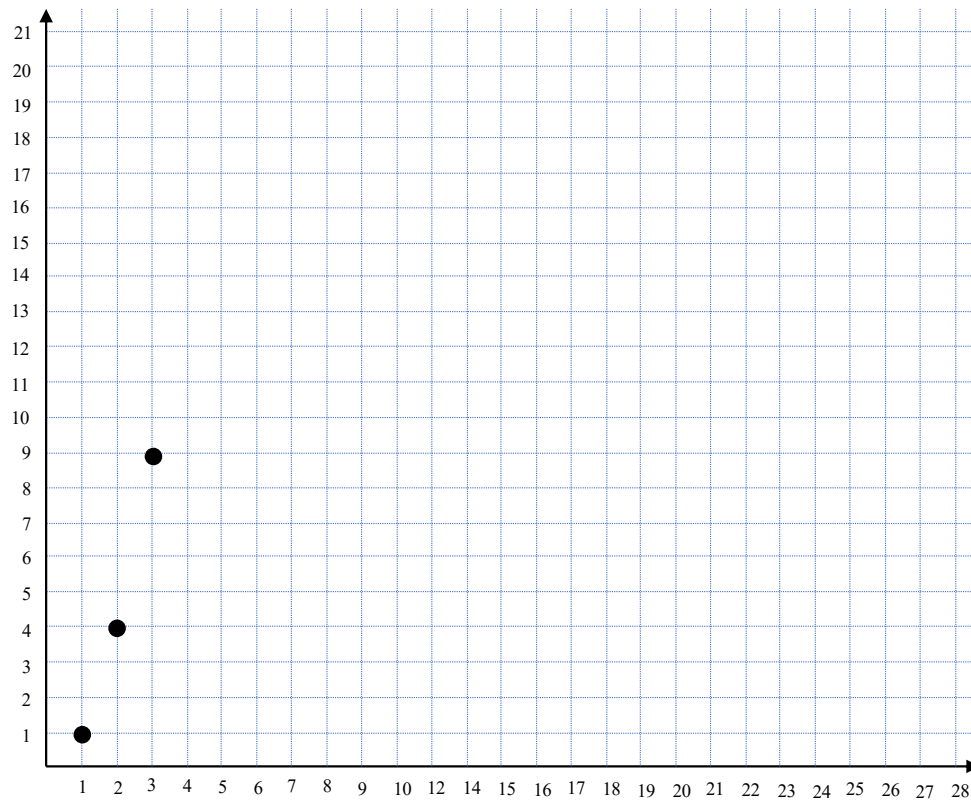
$y = f(x)$



x

x	$y = f(x) = x^2$
1	$1^2 = 1$
2	$2^2 = 4$

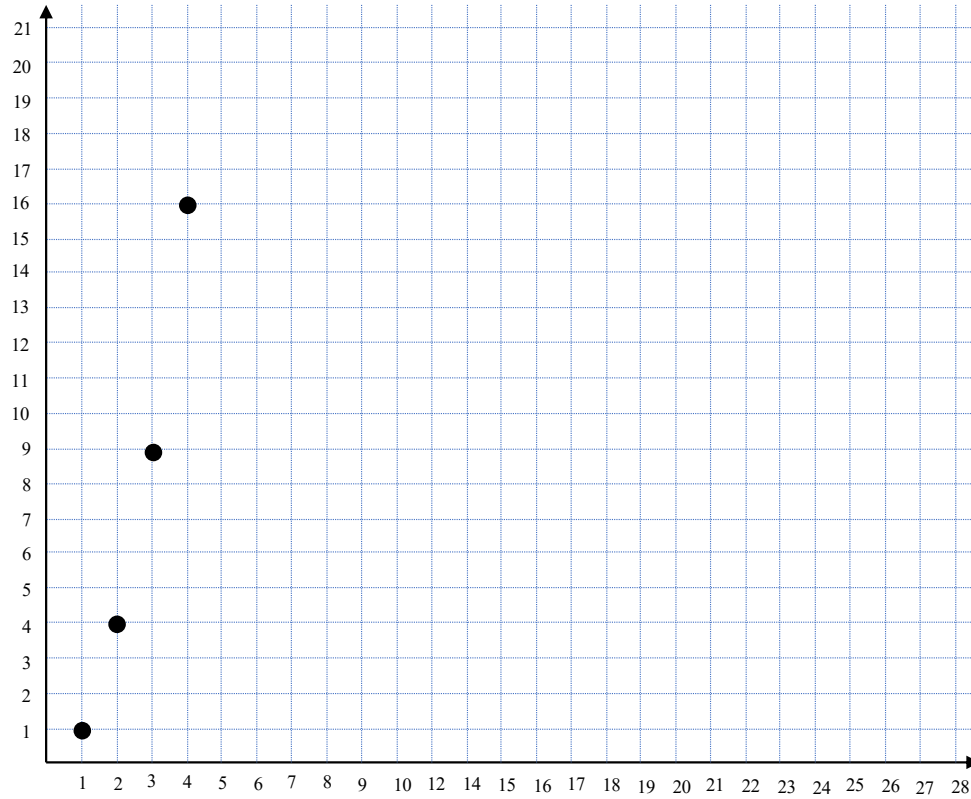
$$y = f(x)$$



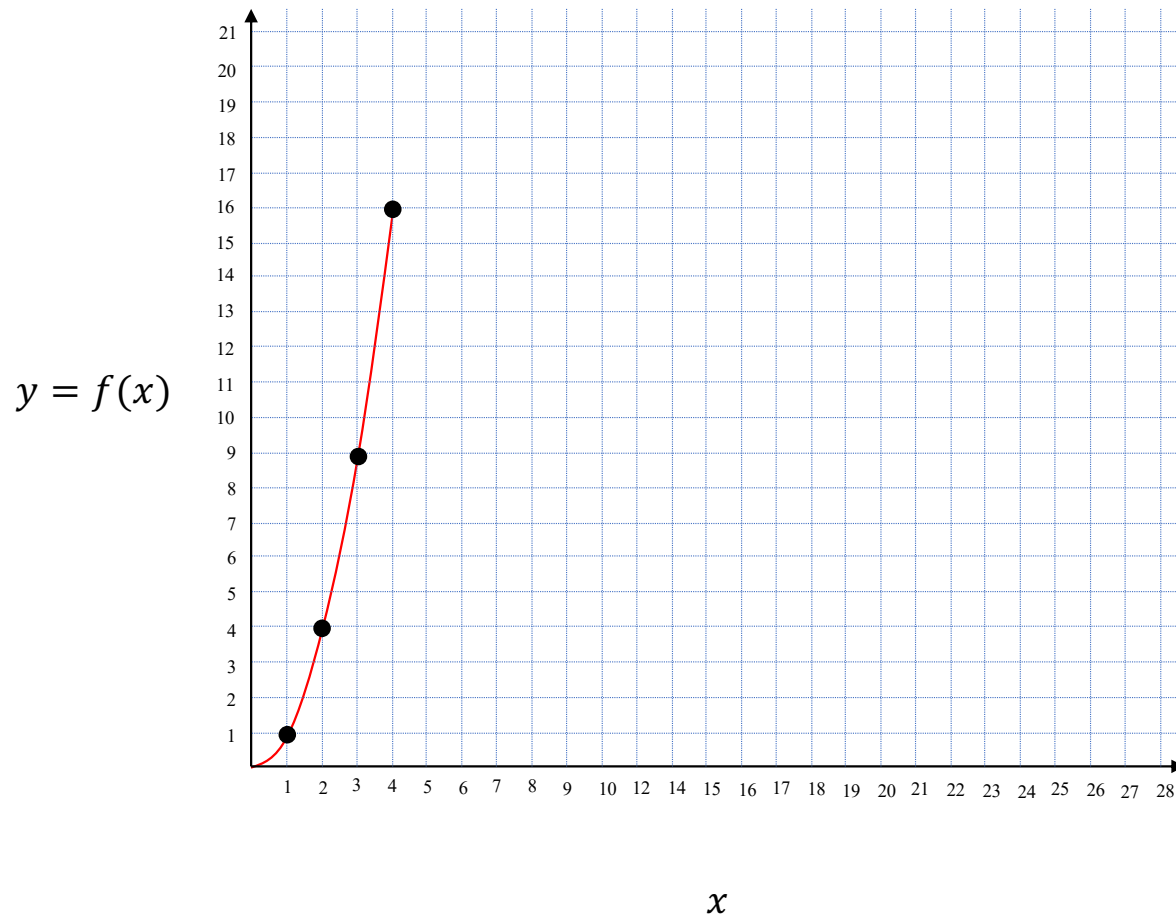
x

x	$y = f(x) = x^2$
1	$1^2 = 1$
2	$2^2 = 4$
3	$3^2 = 9$

$$y = f(x)$$

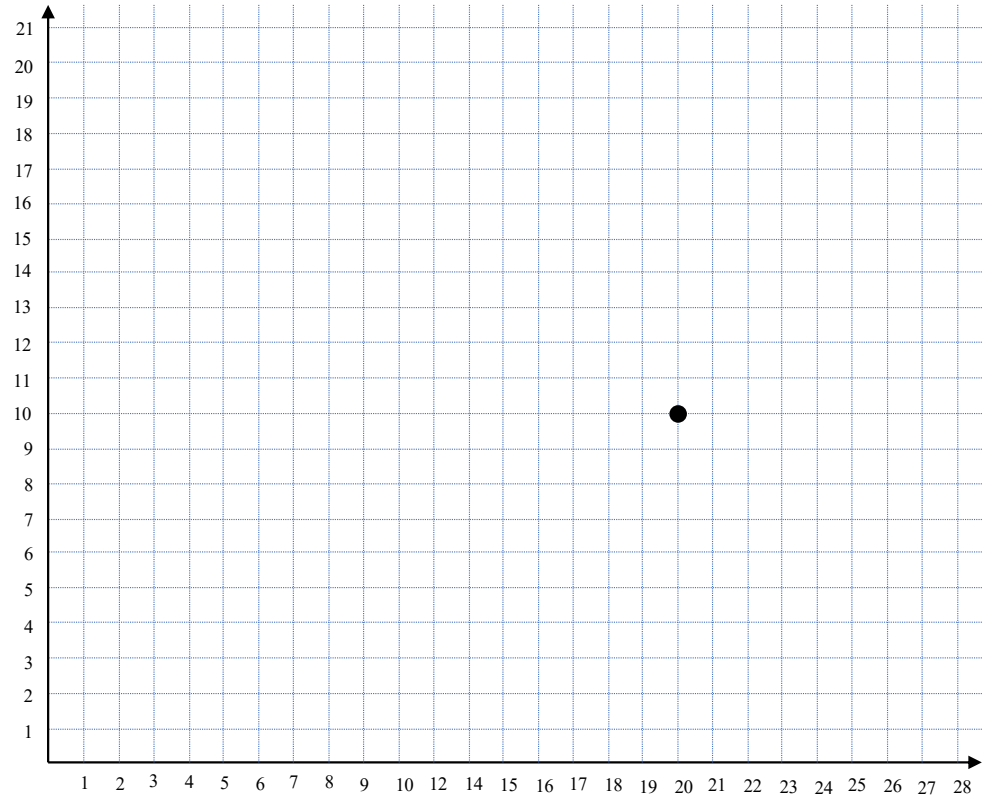


x	$y = f(x) = x^2$
1	$1^2 = 1$
2	$2^2 = 4$
3	$3^2 = 9$
4	$4^2 = 16$



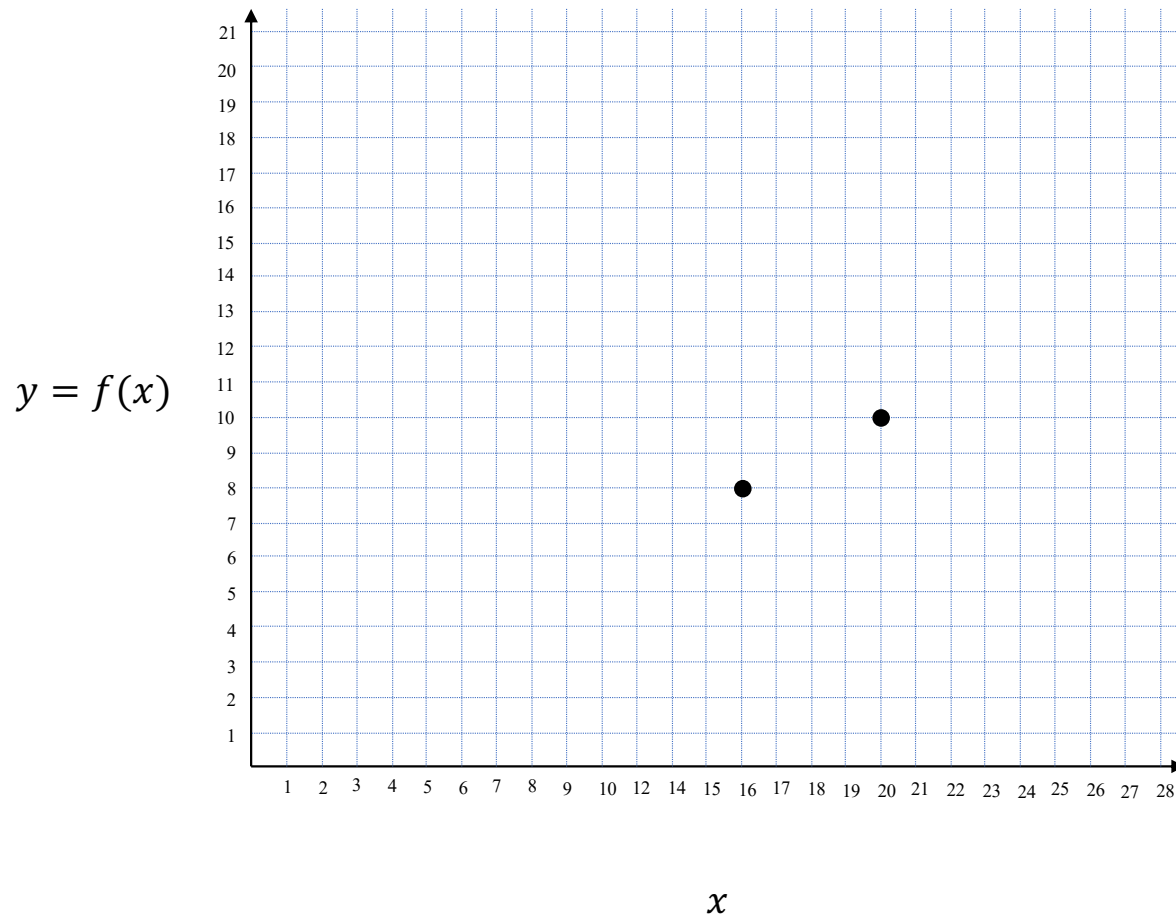
x	$y = f(x) = x^2$
1	$1^2 = 1$
2	$2^2 = 4$
3	$3^2 = 9$
4	$4^2 = 16$

$y = f(x)$

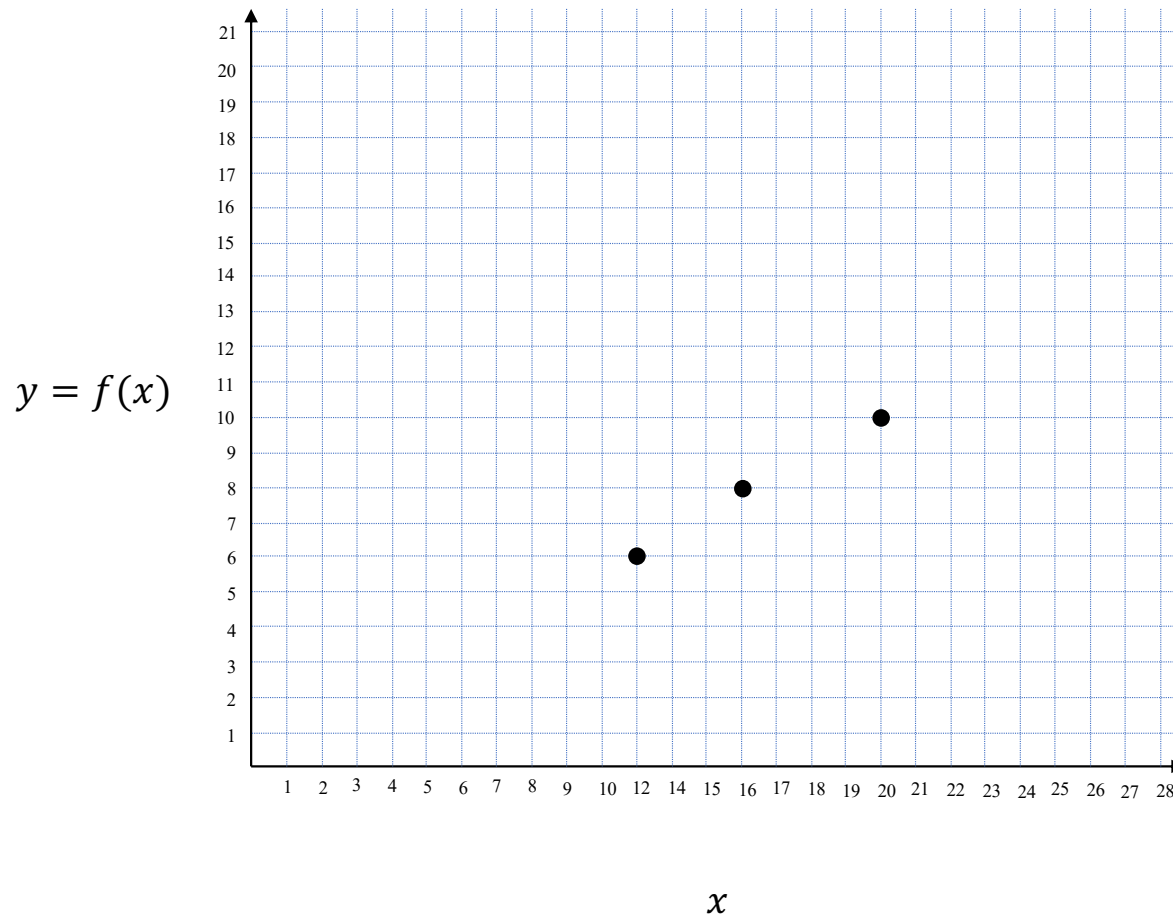


x

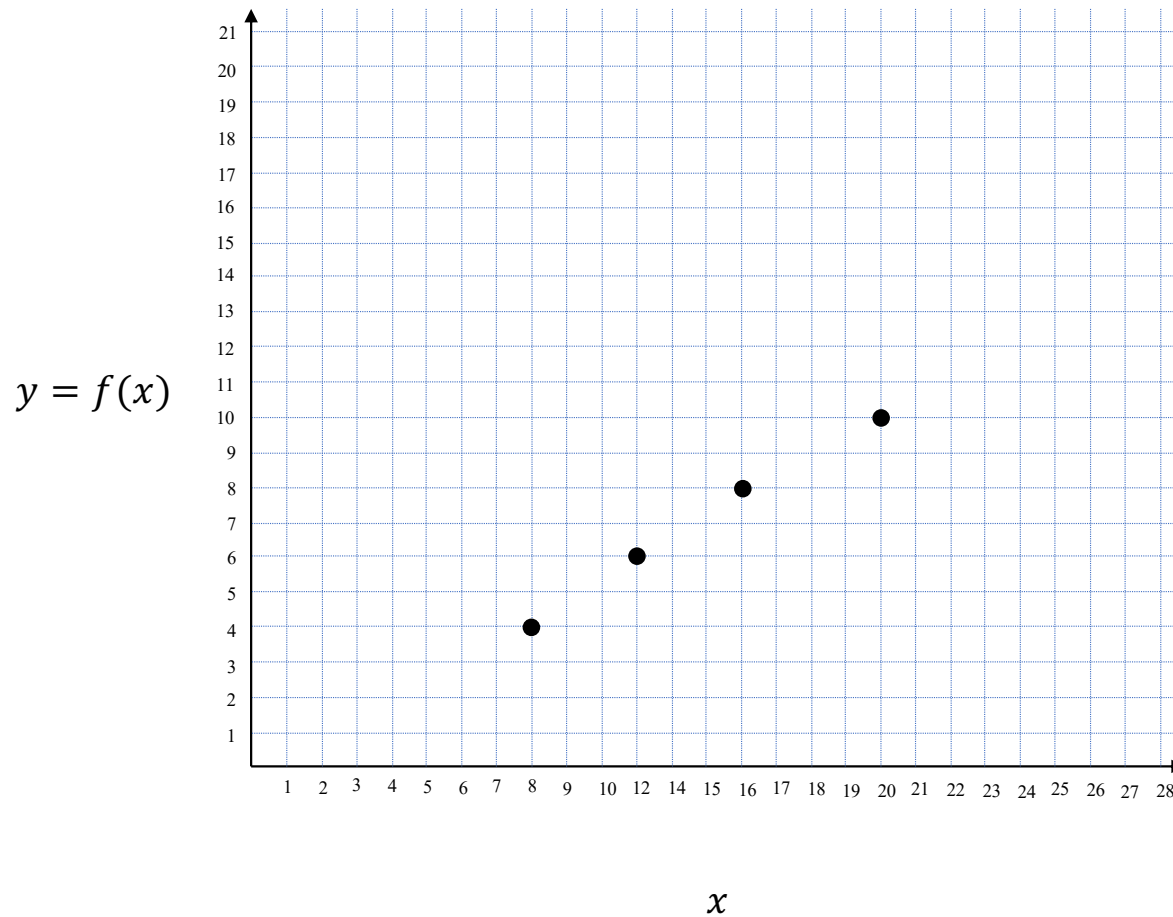
x	$y = f(x) = \frac{x}{2}$
20	$\frac{20}{2} = 10$



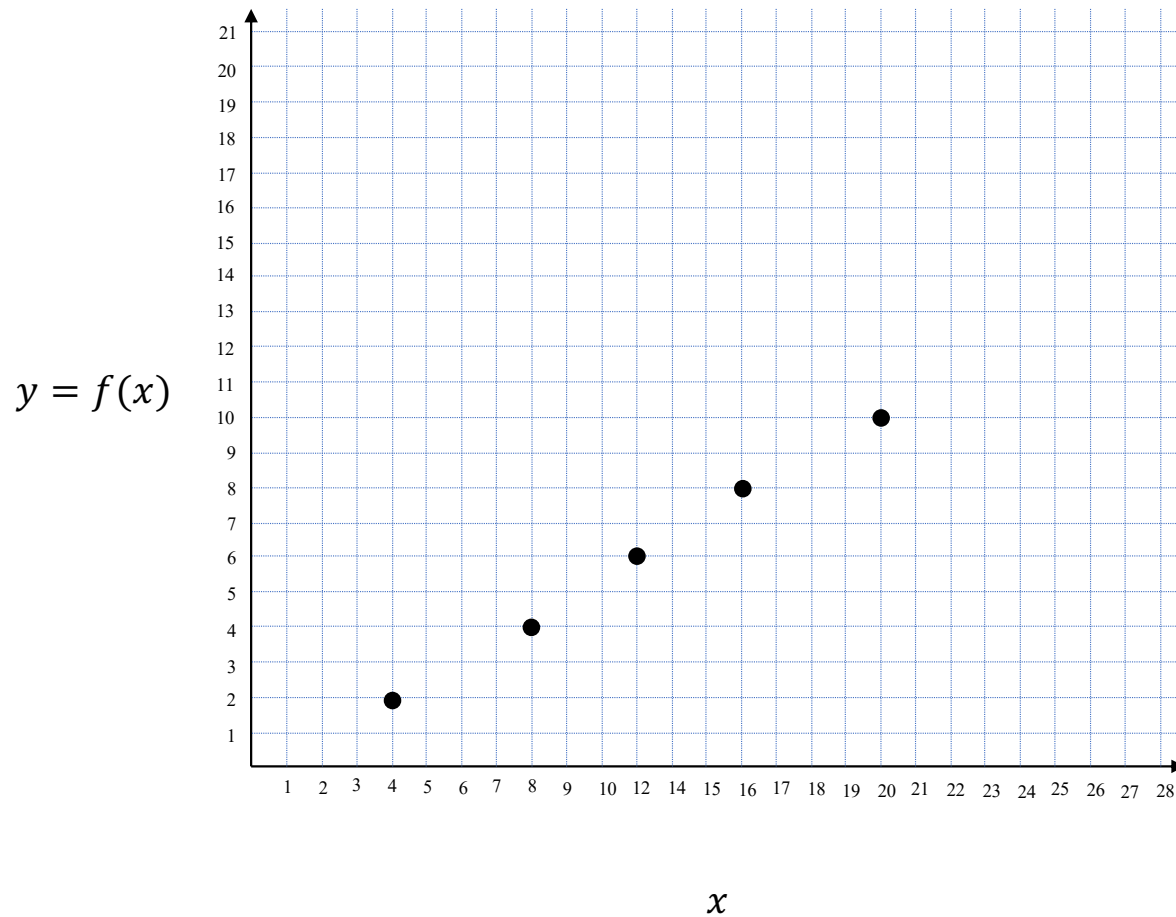
x	$y = f(x) = \frac{x}{2}$
20	$\frac{20}{2} = 10$
16	$\frac{16}{2} = 8$



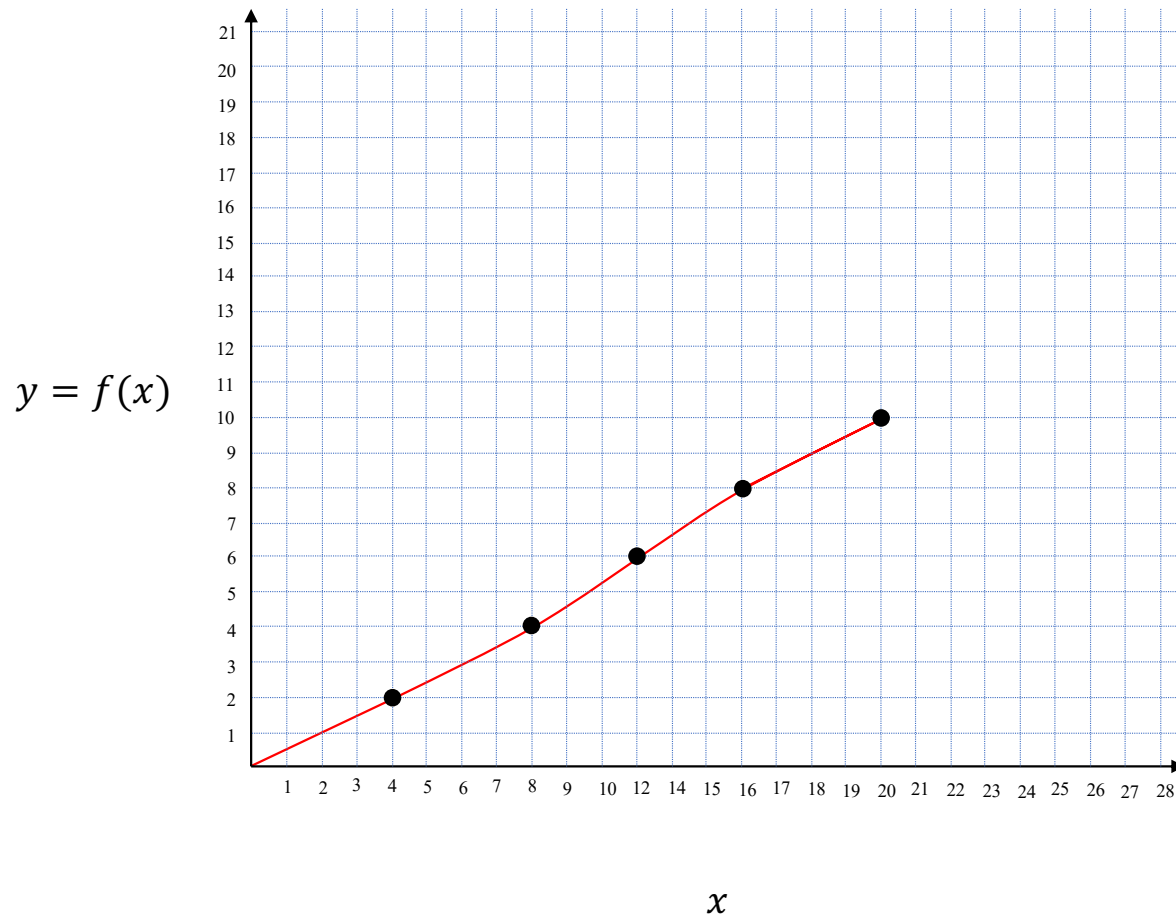
x	$y = f(x) = \frac{x}{2}$
20	$\frac{20}{2} = 10$
16	$\frac{16}{2} = 8$
12	$\frac{12}{2} = 6$



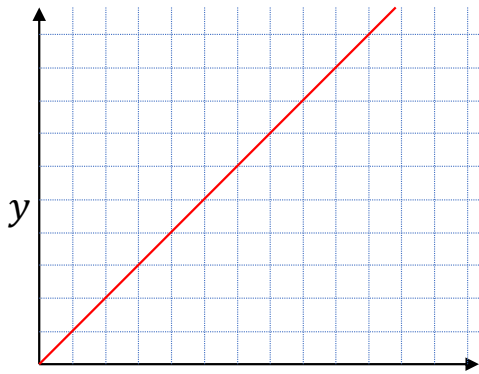
x	$y = f(x) = \frac{x}{2}$
20	$\frac{20}{2} = 10$
16	$\frac{16}{2} = 8$
12	$\frac{12}{2} = 6$
8	$\frac{8}{2} = 4$



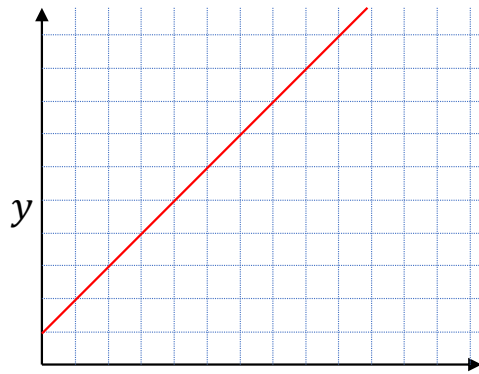
x	$y = f(x) = \frac{x}{2}$
20	$\frac{20}{2} = 10$
16	$\frac{16}{2} = 8$
12	$\frac{12}{2} = 6$
8	$\frac{8}{2} = 4$
4	$\frac{4}{2} = 2$



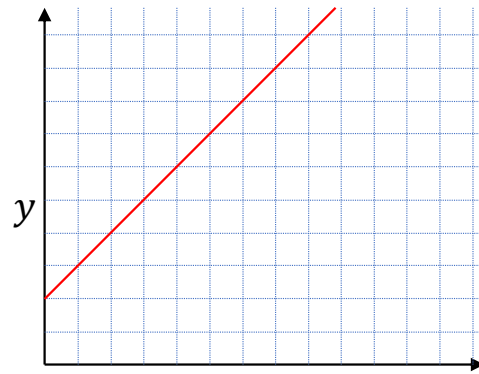
x	$y = f(x) = \frac{x}{2}$
20	$\frac{20}{2} = 10$
16	$\frac{16}{2} = 8$
12	$\frac{12}{2} = 6$
8	$\frac{8}{2} = 4$
4	$\frac{4}{2} = 2$



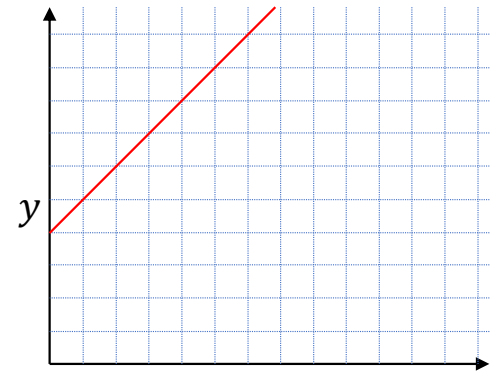
x
 $f(x) = x + 0$



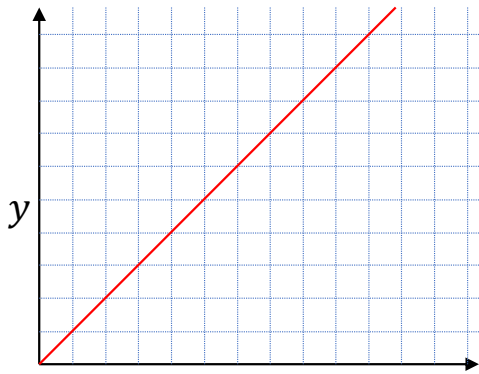
x
 $f(x) = x + 1$



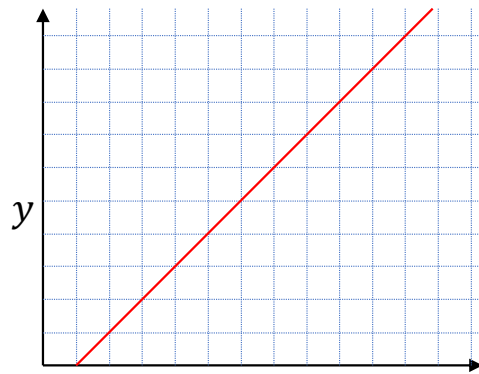
x
 $f(x) = x + 2$



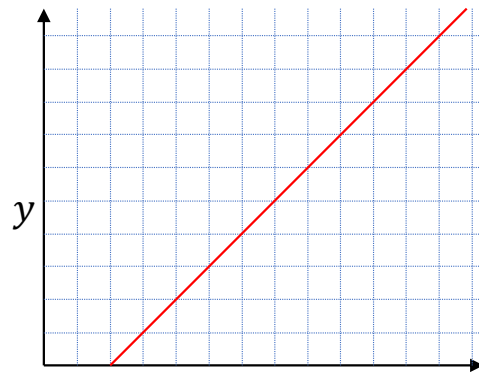
x
 $f(x) = x + 4$



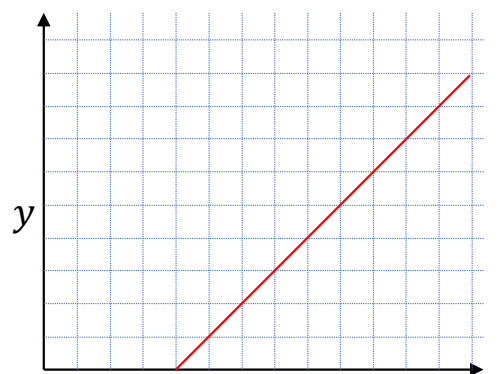
x
 $f(x) = x + 0$



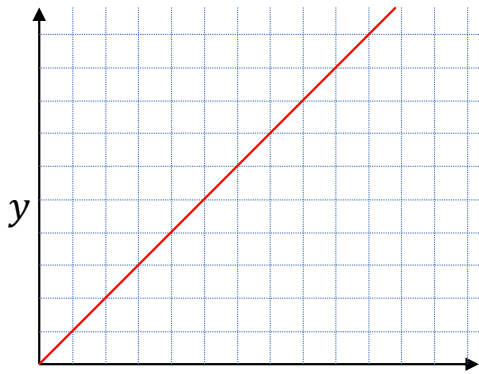
x
 $f(x) = x - 1$



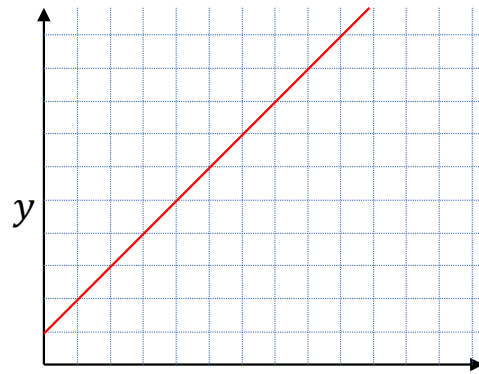
x
 $f(x) = x - 2$



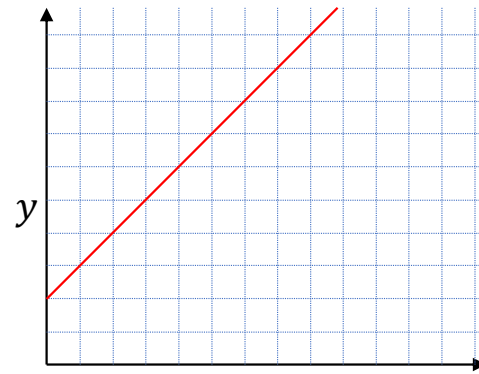
x
 $f(x) = x - 4$



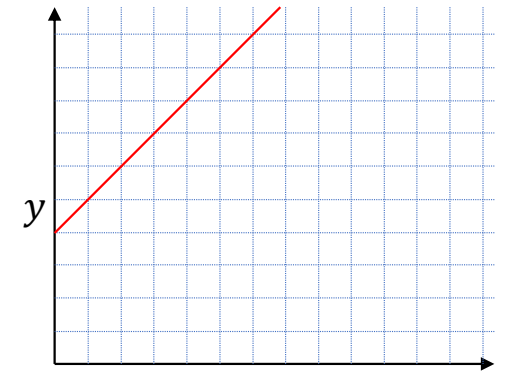
$$f(x) = x + 0$$



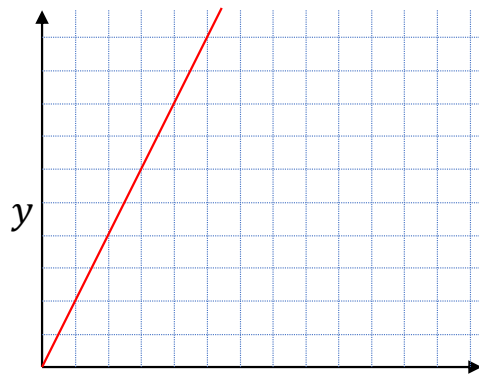
$$f(x) = x + 1$$



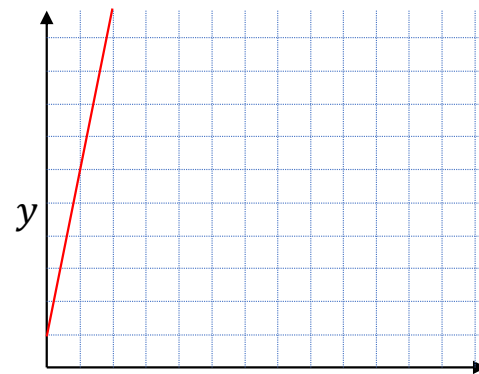
$$f(x) = x + 2$$



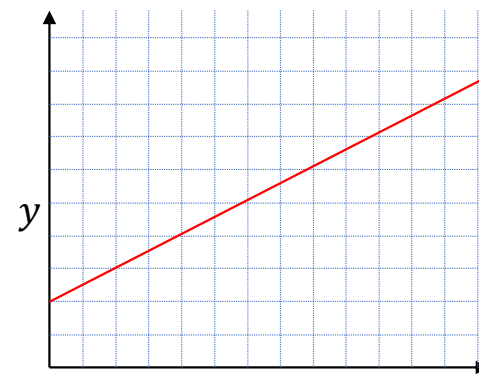
$$f(x) = x + 4$$



$$f(x) = 2x + 0$$



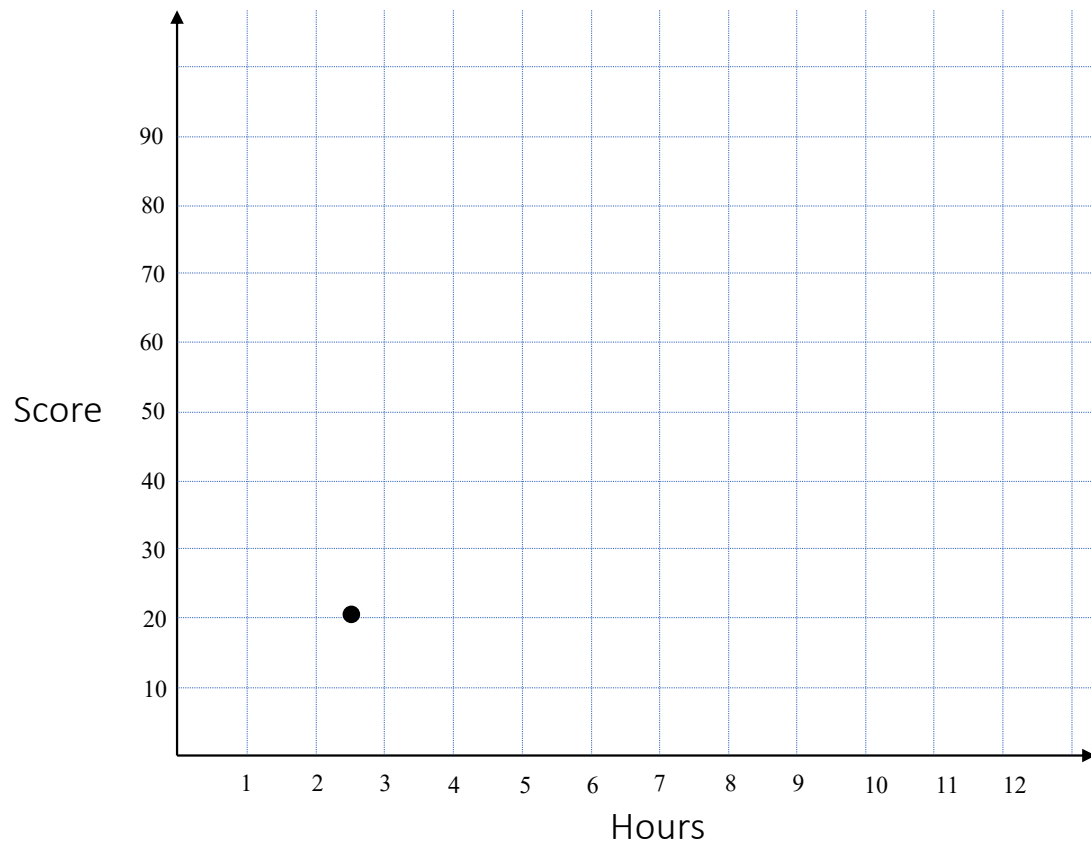
$$f(x) = 5x + 1$$



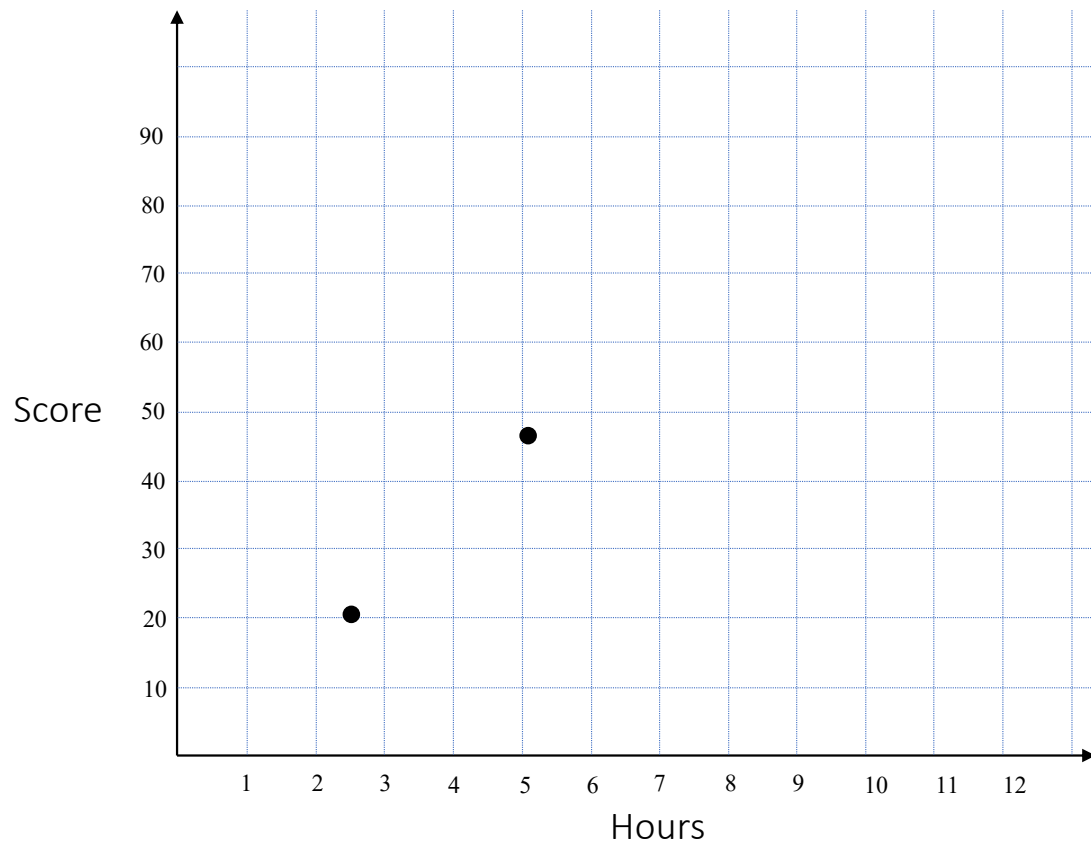
$$f(x) = 0.5x + 2$$



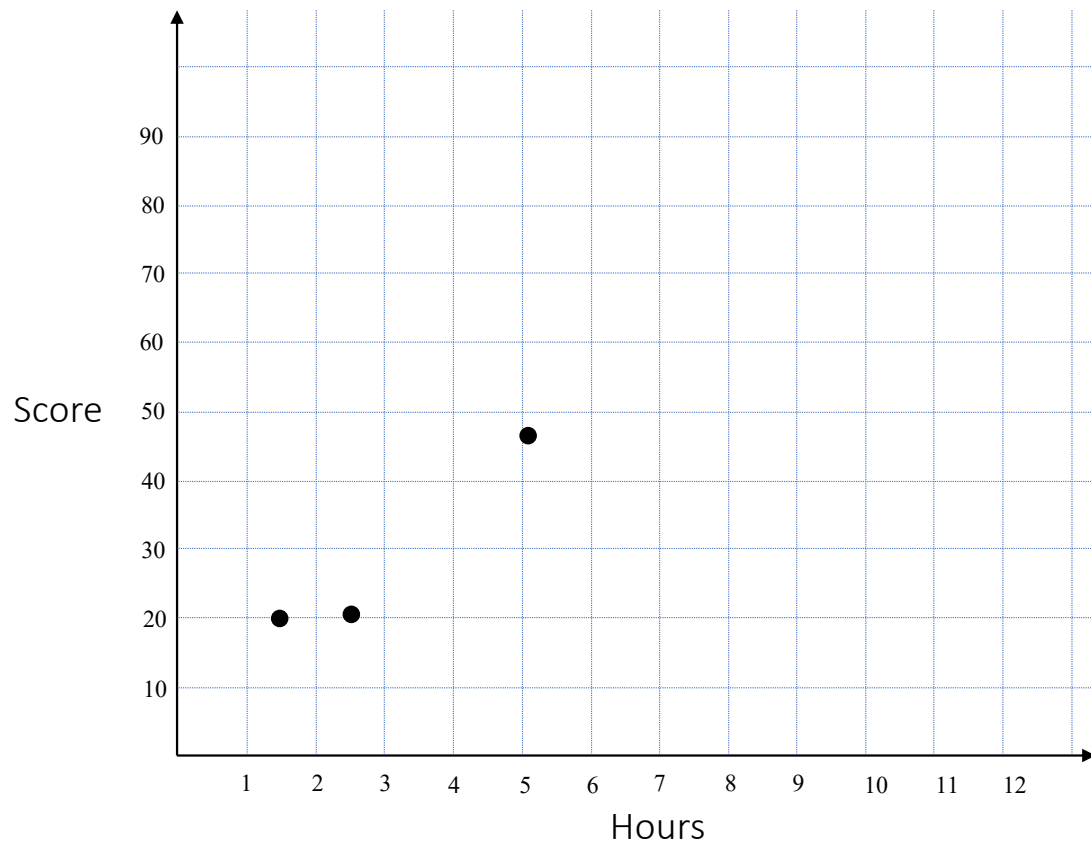
$$f(x) = -1x + 4$$



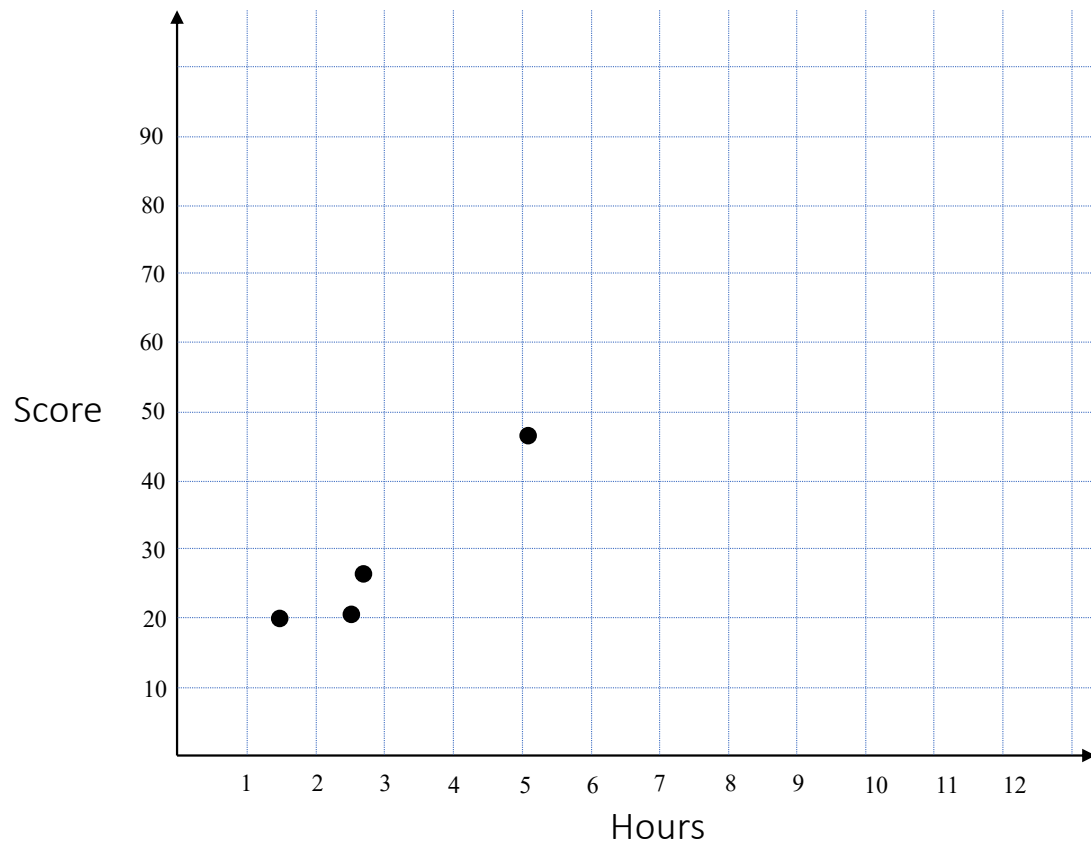
Hour	Score
2.5	21



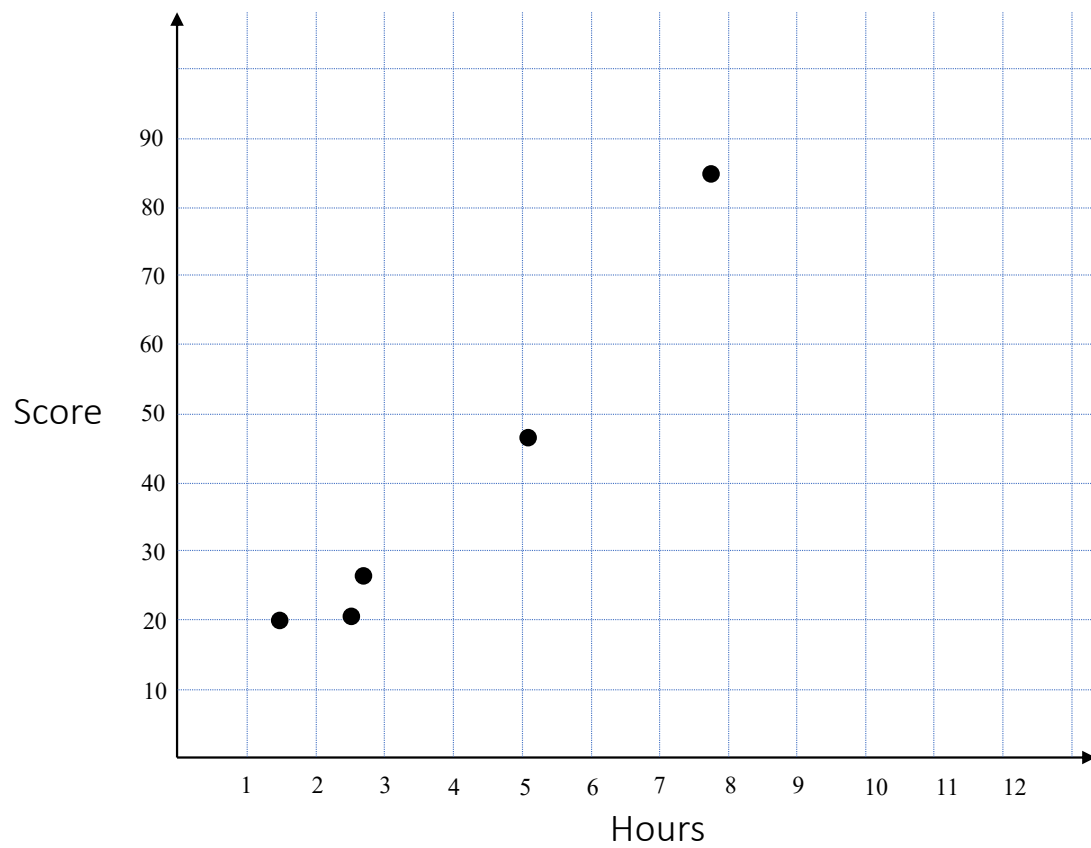
Hour	Score
2.5	21
5.1	47



Hour	Score
2.5	21
5.1	47
1.5	20



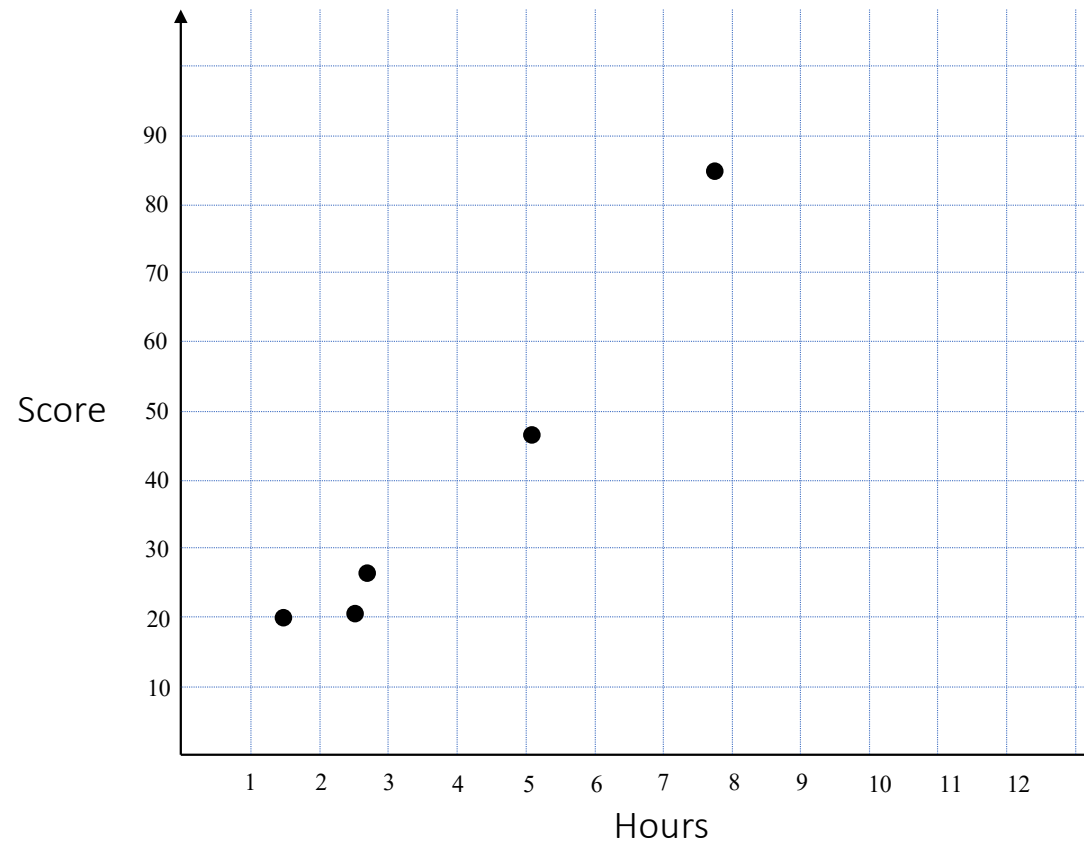
Hour	Score
2.5	21
5.1	47
1.5	20
2.7	25



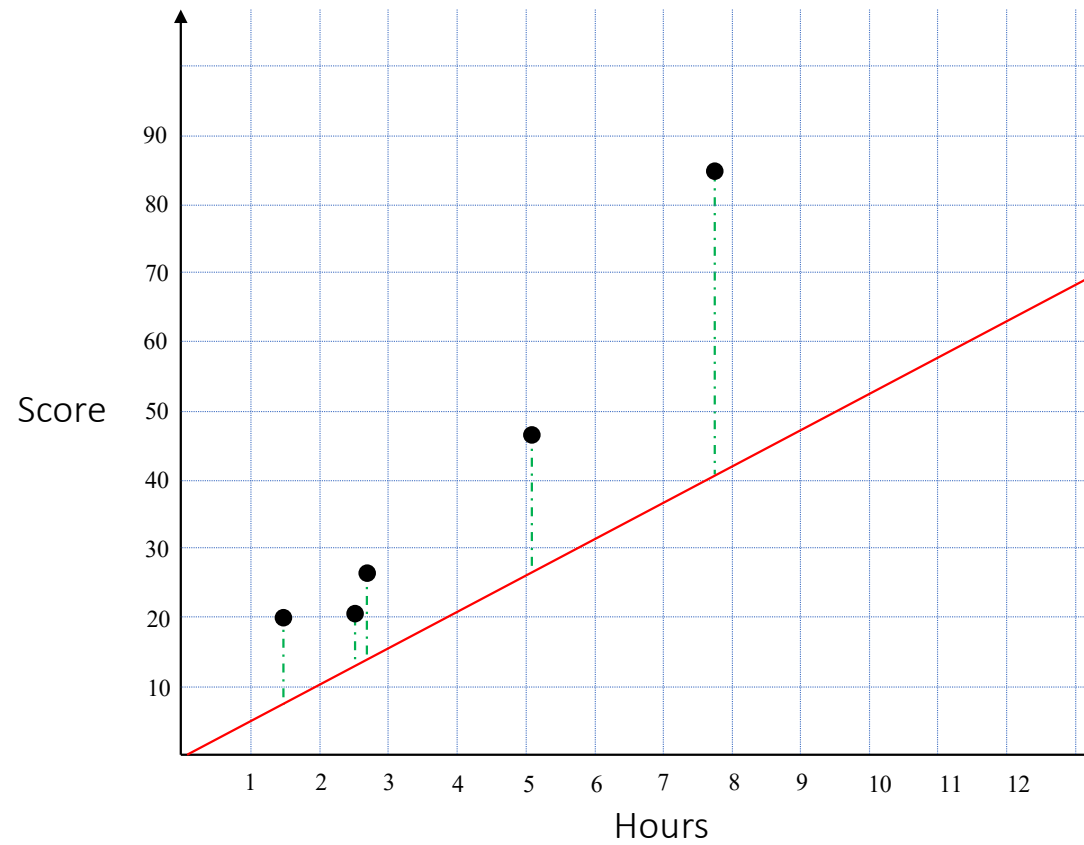
Hour	Score
2.5	21
5.1	47
1.5	20
2.7	25
7.7	85

Question: Now what?

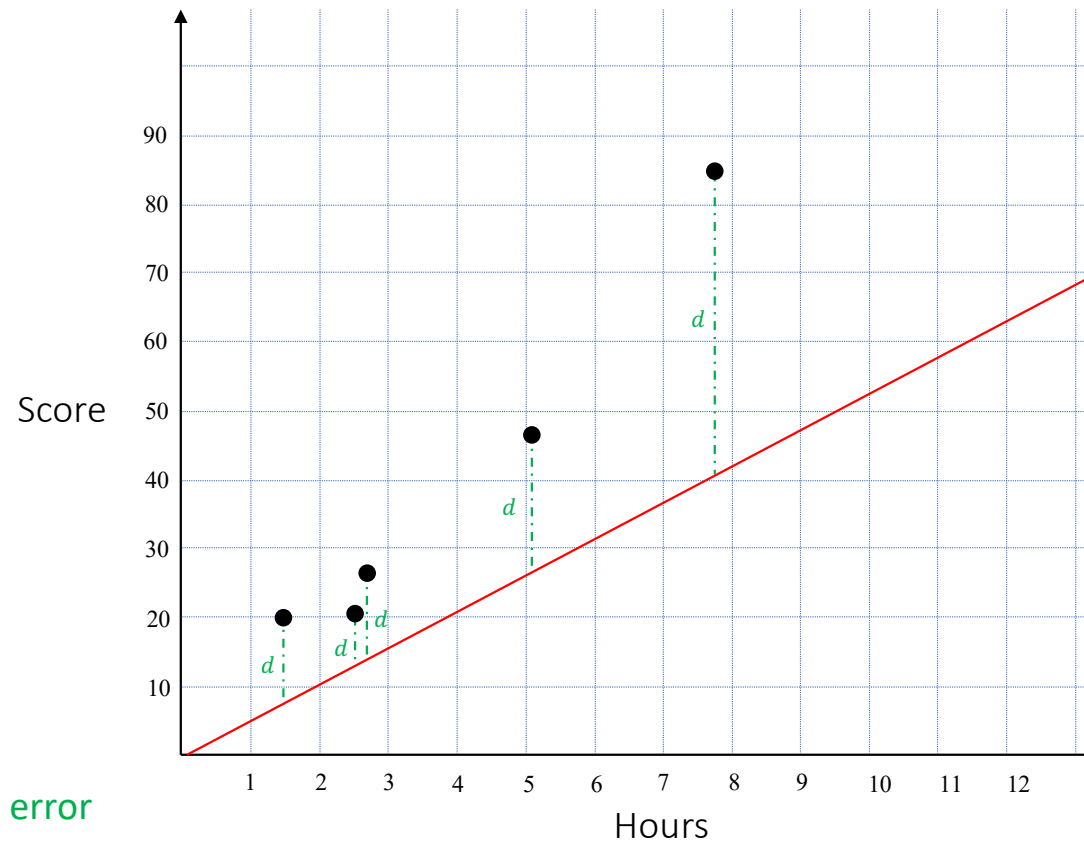
Answer: Line fitting.



Let's draw a random line

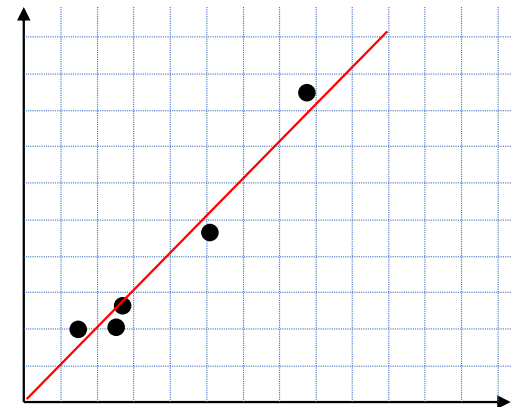
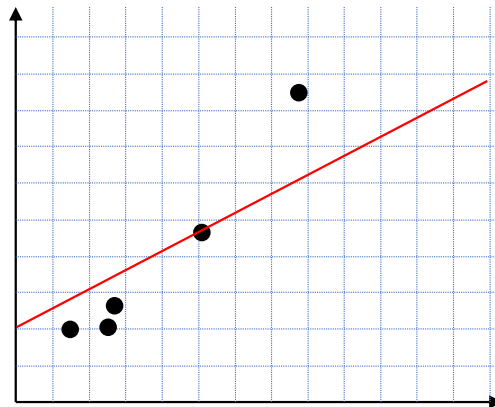
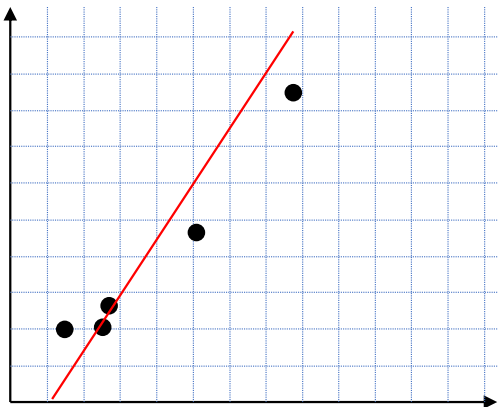
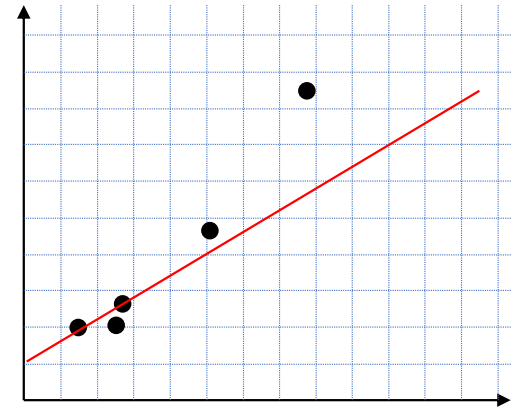
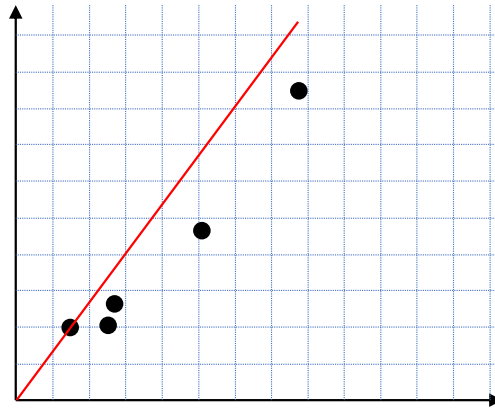
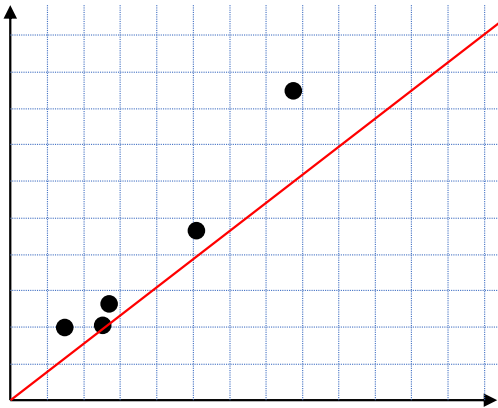


Let's draw a random line



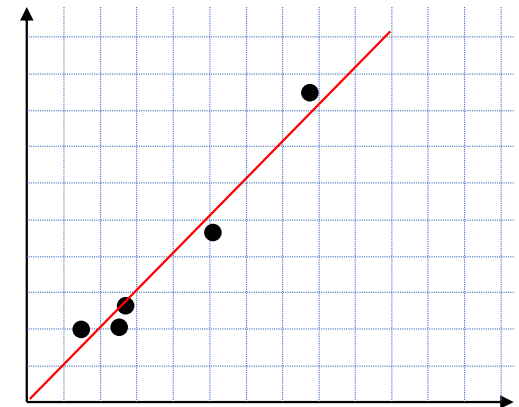
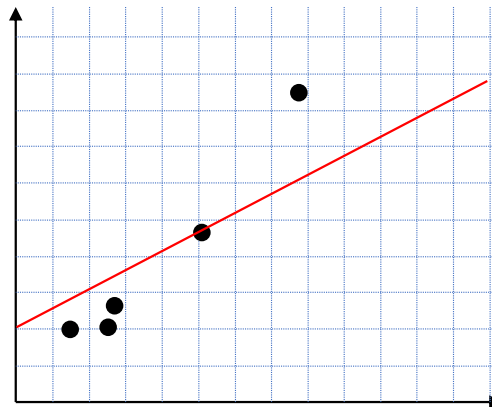
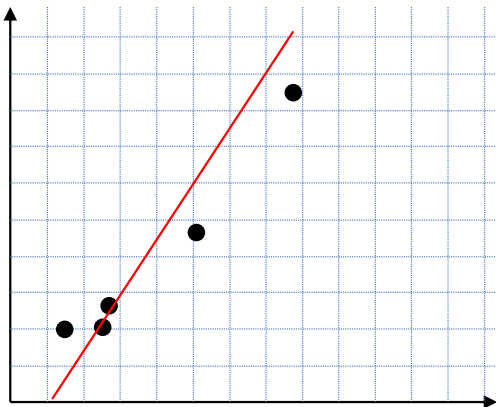
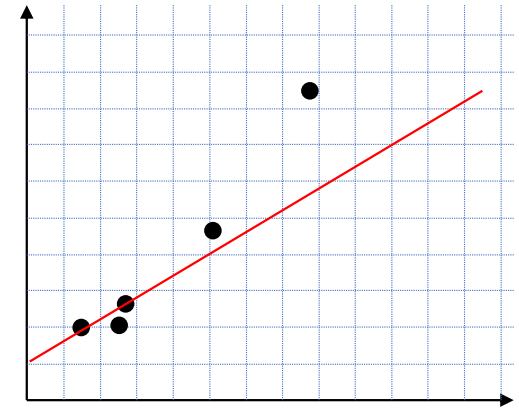
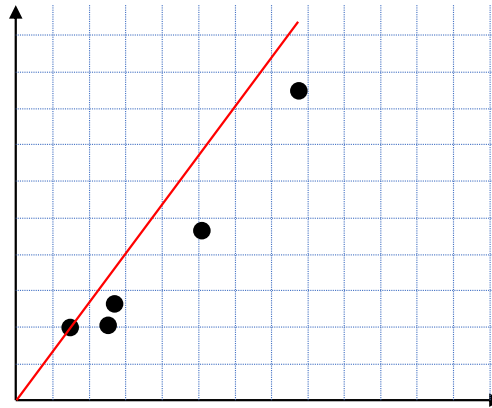
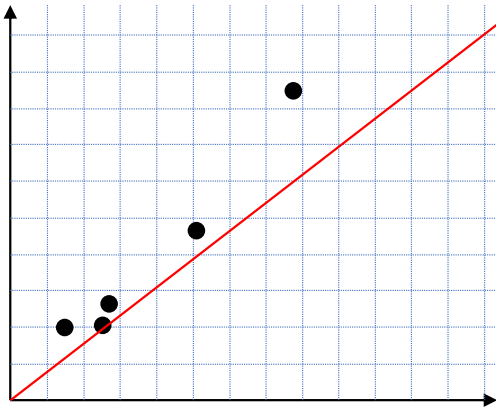
Green color represents error

Which one is best fitting the data-points?

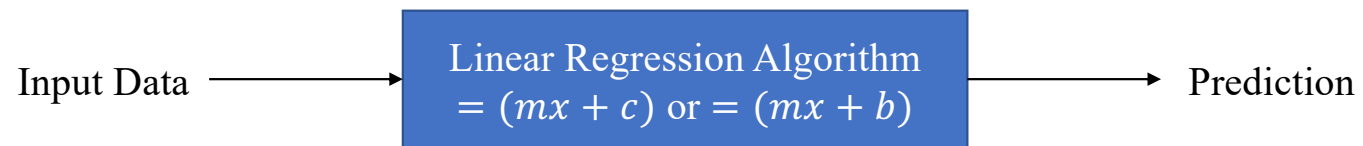


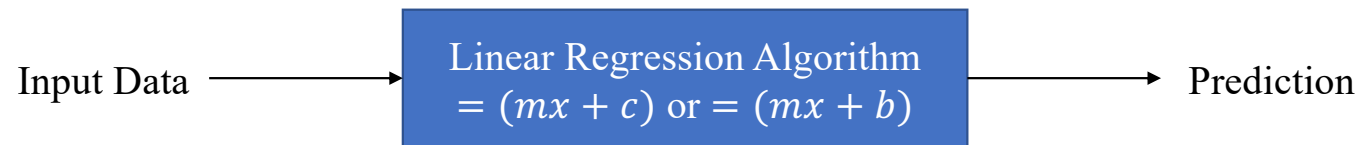
Which one is best fitting the data-points?

The last one.

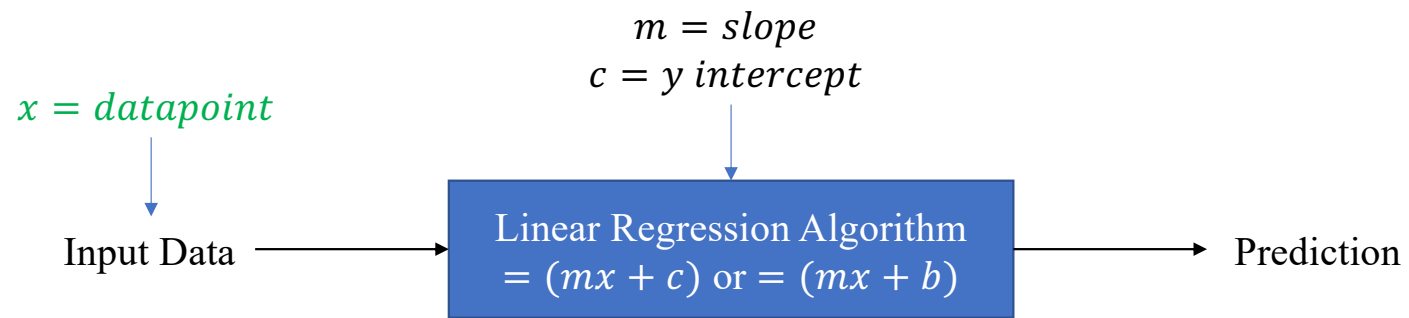


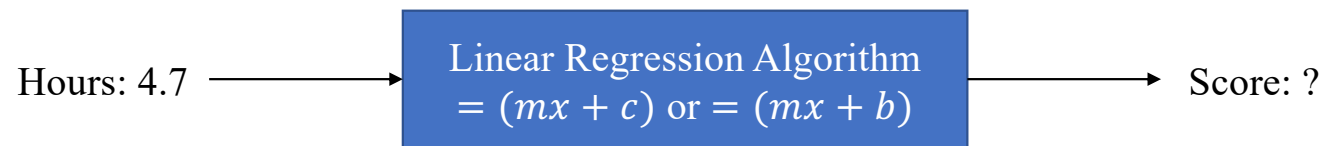
Before coding, let's put up things together



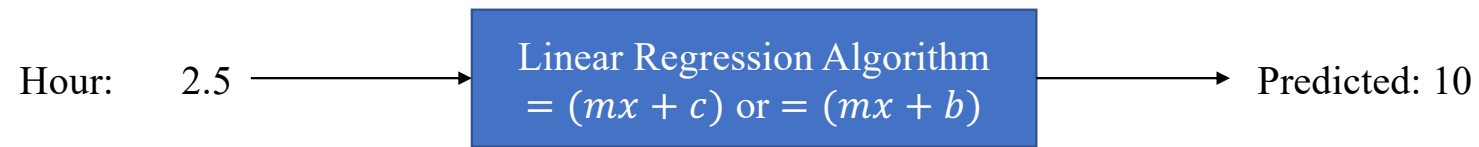


$m = \text{slope}$
 $x = \text{datapoint}$
 $c = y \text{ intercept}$

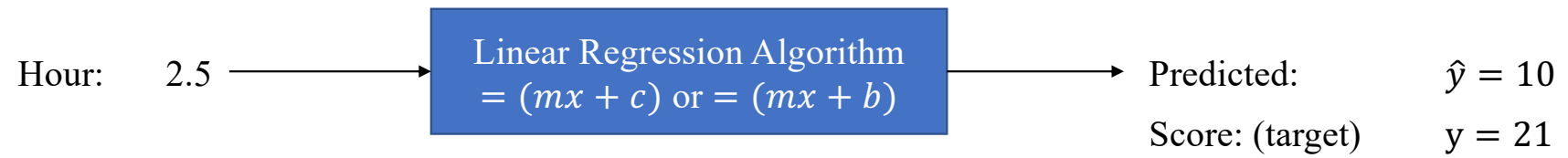




Hour	Score
2.5	21
5.1	47
1.5	20
2.7	25
7.7	85



Hour	Score
2.5	21
5.1	47
1.5	20
2.7	25
7.7	85



Predicted: $\hat{y} = 10$

Score: (target) $y = 21$

Simple Error	Square Error	Mean Square Error
$e = (y - \hat{y})$ $e = (21 - 10)$ $e = 11$	$e = (y - \hat{y})^2$ $e = (21 - 10)^2$ $e = (11)^2 = 121$	$e = \frac{1}{N} \sum (y - \hat{y})^2$

or

$$e = \sum (y - \hat{y})$$

For multiple datapoints

or

$$e = \sum (y - \hat{y})^2$$

For multiple datapoints

We will be using this one

Hour	Score
2.5	21
5.1	47
1.5	20
2.7	25
7.7	85

$$e = \frac{1}{N} \sum (y - \hat{y})^2$$

$$e = \frac{1}{5} ((21 - 10)^2 + (47 - 12)^2 + (20 - 8)^2 + (25 - 11.5)^2 + (85 - 41)^2)$$

$$e = \frac{1}{5} (11^2 + 35^2 + 12^2 + 13.5^2 + 44^2)$$

$$e = \frac{1}{5} (121 + 1225 + 144 + 182.25 + 1936)$$

$$e = \frac{1}{5} (3608.25)$$

$$e = 721.65$$

$x = \text{datapoint}$

Hour

$m = \text{slope}$
 $c = y \text{ intercept}$

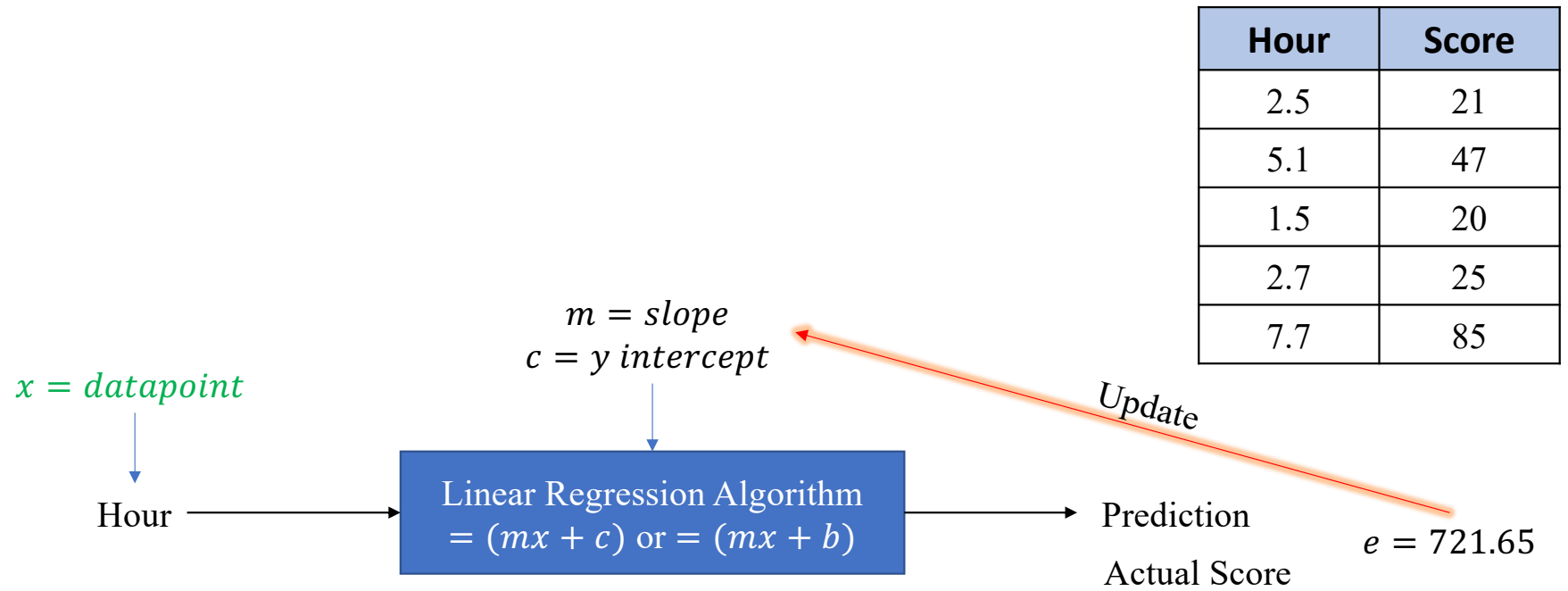
Linear Regression Algorithm
 $= (mx + c) \text{ or } = (mx + b)$

Prediction

Actual Score

$e = 721.65$

Hour	Score
2.5	21
5.1	47
1.5	20
2.7	25
7.7	85



$$m = 0$$

$$c = 0$$

To update these weights...

$$\textit{New Weight} = \textit{Old Weight} - \textit{Learning Rate}(\textit{Error})$$

$$e = 721.65$$

$$m = 0$$

$$c = 0$$

To update these weights...

$$\text{New Weight} = \text{Old Weight} - \text{Learning Rate}(\text{Error})$$

$$\text{Let Learning Rate} = 0.0001$$

$$m = m - lr(e)$$

$$e = 721.65$$

$$m = 0 - 0.0001(721.65)$$

$$m = 0.072$$

$$c = c - lr(e)$$

$$c = 0 - 0.0001(721.65)$$

$$c = 0.072$$

$$m = 0$$

$$c = 0$$

To update these weights...

$$\text{New Weight} = \text{Old Weight} - \text{Learning Rate}(\text{Error})$$

$$\text{Let Learning Rate} = 0.0001$$

$$m = m - lr(e)$$

$$e = 721.65$$

$$m = 0 - 0.0001(721.65)$$

$$m = 0.072$$

$$c = c - lr(e)$$

$$c = 0 - 0.0001(721.65)$$

$$c = 0.072$$

$$m = 0.072$$

$$c = 0.072$$

- The error, best known as Gradient Descent in Linear Regression.

$$e = \frac{1}{N} \sum (y - \hat{y})^2$$

- The formula required to update or tune weights of the model.

$$\bar{w} = w - \alpha (e)$$

Let's Code